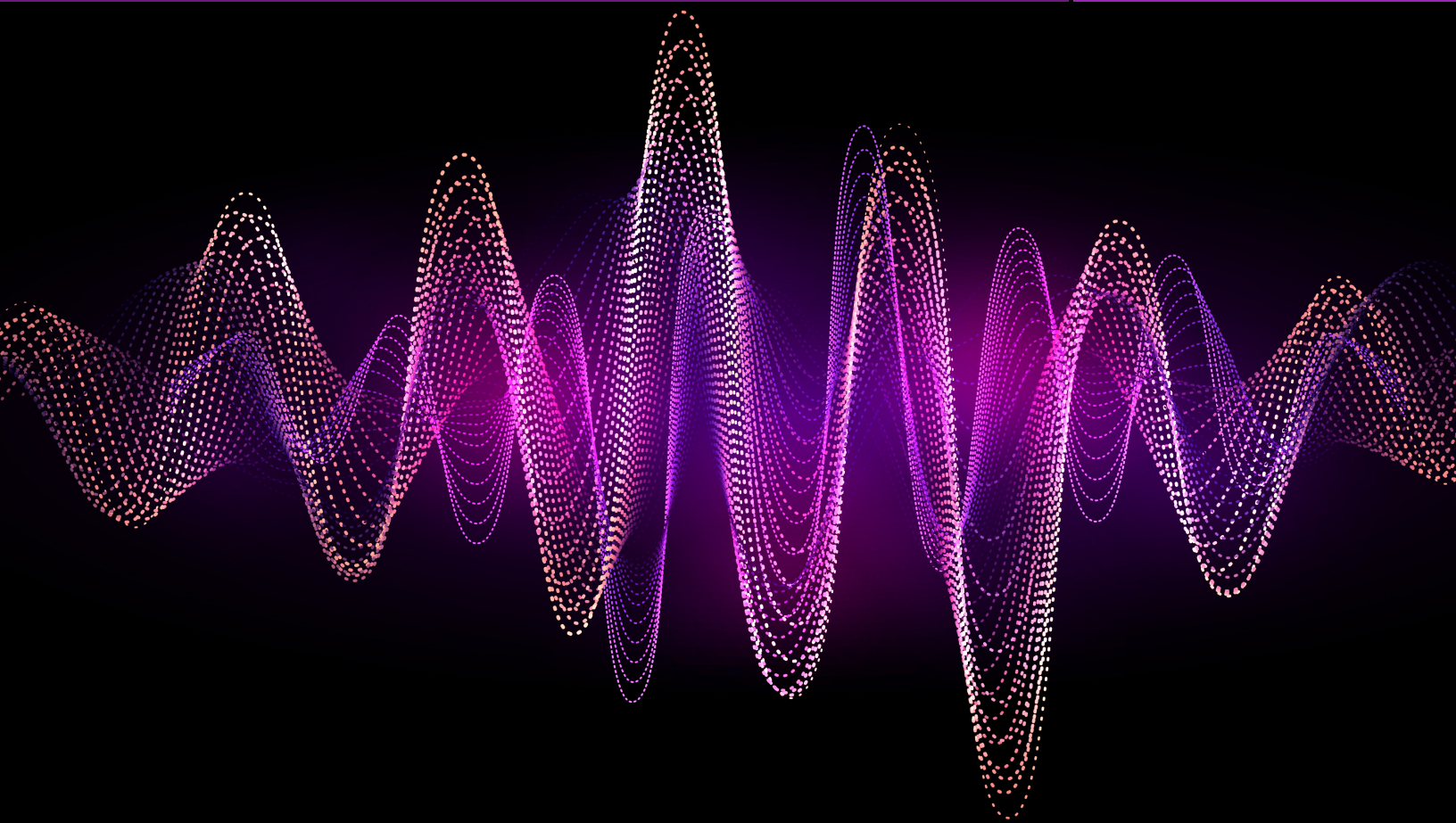




Everything You Need to Know About Sample Rate

Oscilloscope signal integrity: Part 2

Introduction

As our world becomes ever more dependent on technology, engineers are feeling the increased pressure to develop faster and more efficient devices. When designing a new device, the last thing you want is a redesign and delay due to an error that should have been caught earlier on in development.

Oscilloscopes are well-known instruments. However, getting a useful, accurate view of your signal isn't as simple as plugging it in, turning it on, and connecting the probe to your device. One of the key specifications that dramatically impacts the signal you see on the display is the sample rate. Using an incorrect sample rate or acquisition mode can result in an inaccurate or distorted representation of your signal.

To accurately test your design, it is critical to know what sample rate is, why you should care about it, and how to ensure you are using the proper sample rate.

Effective number of bits (ENOB) was established by IEEE in 1993 as a measurement of an oscilloscope's signal integrity and accuracy.

What is Sample Rate?

Modern oscilloscopes convert your analog signal input into a digital signal so that you can see it on the display. The conversion works by taking data points, or samples, of your analog signal and using those points to reconstruct the waveform digitally.

You can see the path that your signal takes through the oscilloscope in Figure 1. Once your signal reaches the front end of the oscilloscope, the sampler will collect data points. Those data points feed into the analog to digital converter (ADC) where they are converted to digital values. They are stored in memory and eventually processed then displayed on the screen.

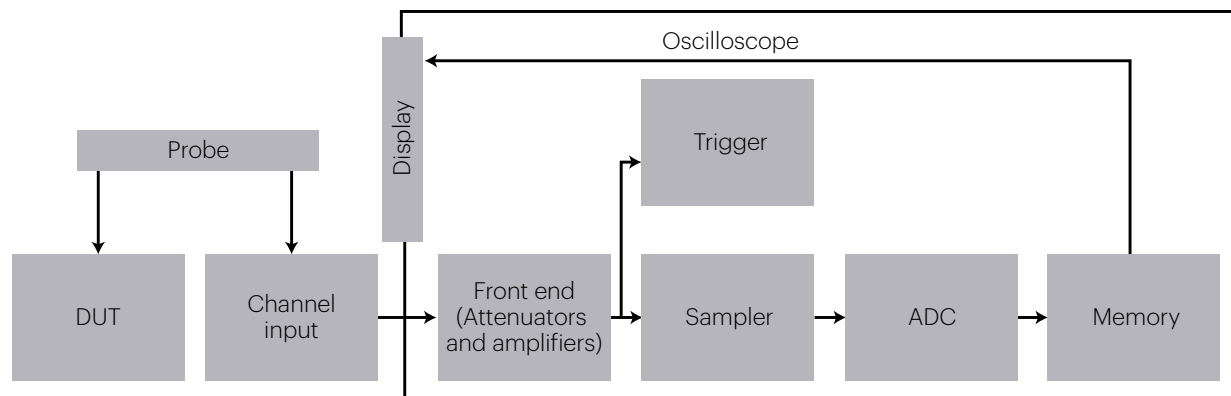


Figure 1. Diagram showing signal path

Simply put, the sample rate is just the rate at which the sampler collects samples. The sample rate is measured in samples per second (Sa/s). This rate ultimately affects the signal you see on screen. In a well-designed oscilloscope, samples will occur at a consistent rate with no variation. Together, the samples collected depict the waveform you see on the oscilloscope display. You see a smooth waveform on the oscilloscope screen rather than sample “dots” you see in Figure 2. This is because the oscilloscope will interpolate the trace between samples, like a game of connect-the-dots.

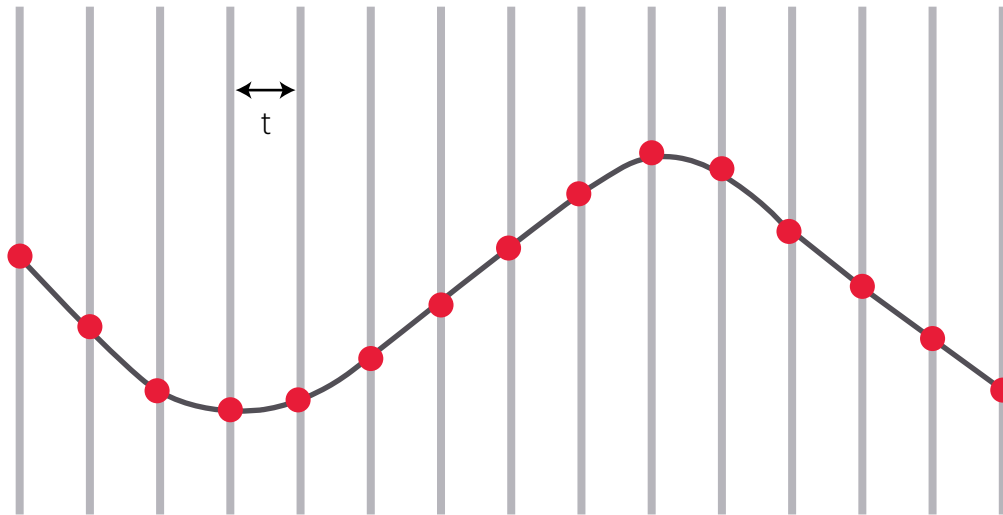


Figure 2. Visualization of the rate at which each sample point is collected

It is important to remember that memory and sample rate are closely linked. As you increase the timebase, more of your waveform will display. This means that more memory is used to store that long timespan of data points. This will effectively decrease the sample rate because the oscilloscope can only store a certain number of sample points in memory. When you adjust the timebase, notice how it changes the sample rate. Too low of a sample rate makes your signal look choppy, cause you to miss some signal detail, and skew your measurements. Always be sure to use the correct scaling for the analysis you are performing.

Why is Sample Rate Important?

In many cases, you have to capture an extended period of data and zoom in on various portions to analyze them further. However, as you learned, significantly increasing the timebase will lower your sample rate. With that low sample rate, you are likely to see a distorted version of your signal, as in Figure 3. You can compare this to the way that a low-resolution digital photo becomes pixelated as you zoom in. This happens because there are not enough samples on the zoomed in portion of the signal to accurately represent the smooth edges. If you were to perform a peak to peak measurement on this, your results would be inaccurate.

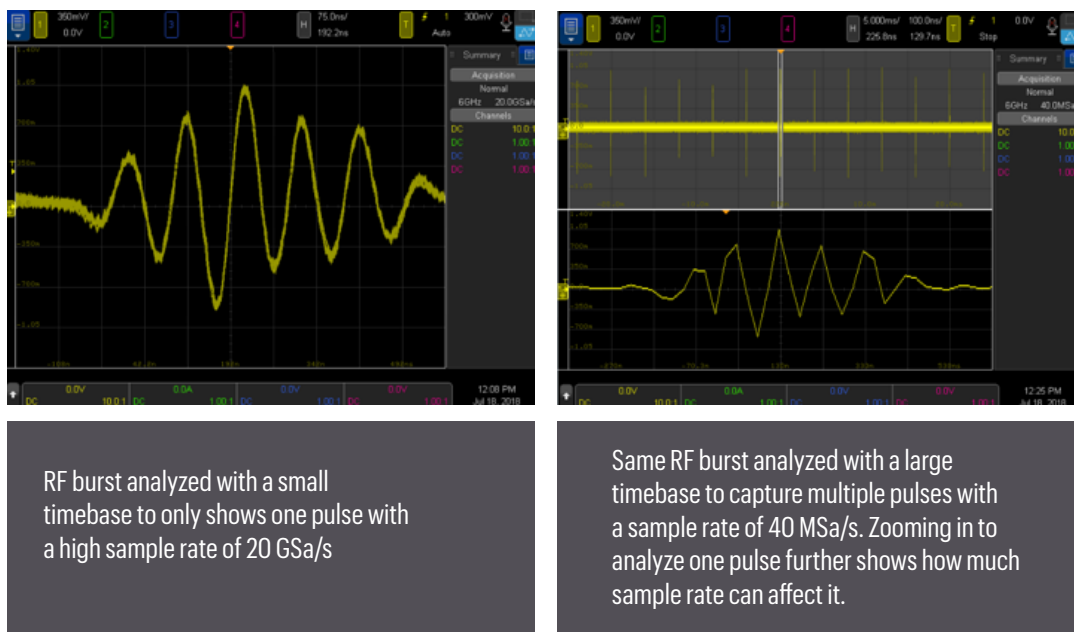


Figure 3. Comparison of an adequately sampled RF burst (left) and an undersampled burst (right) due to change in the timebase, which automatically adjusts the sample rate

Additionally, if your sample rate isn't high enough, you will miss glitches or errors in your signal because elements can occur between collected samples. Figures 4 and 5 illustrate the difference between undersampling and a proper sampling rate. Since glitches and spurs can mean the difference between a device functioning correctly or not, it's critical to detect them in the design phase.

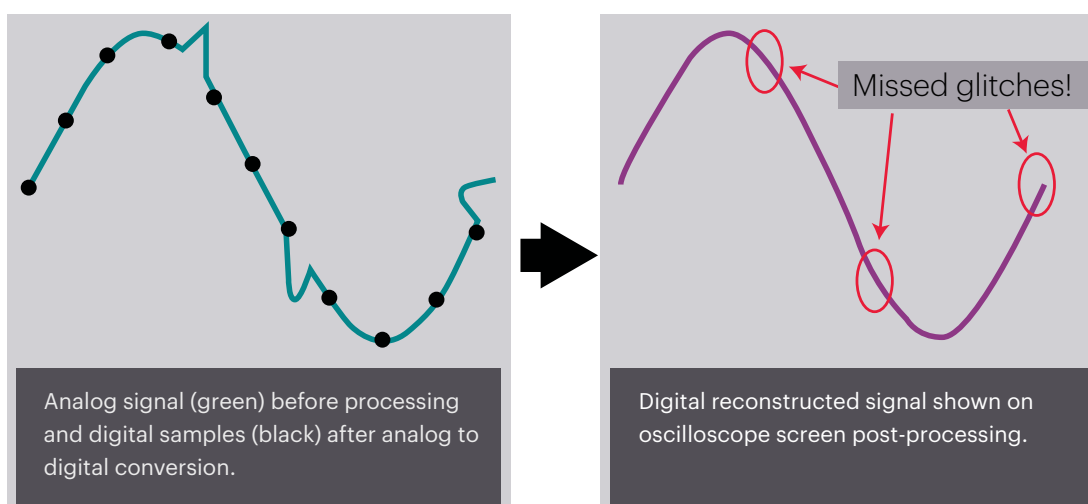


Figure 4. Illustration of missed glitches due to undersampling

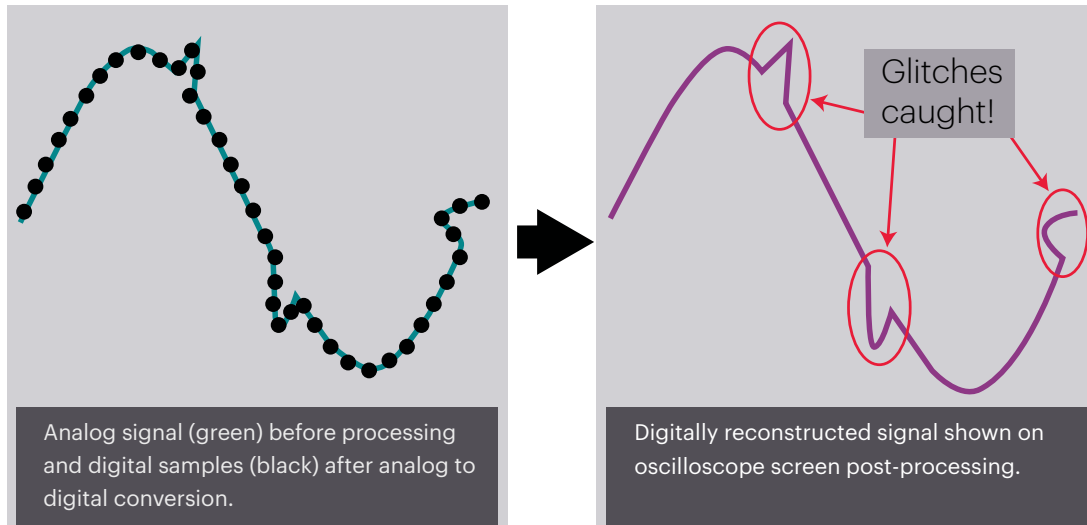


Figure 5. Visualization of a proper, high sample rate accurately capturing glitches

Undersampling could easily cause you to think that your signal is clean, when in fact it has several issues. **Key point: the higher the sample rate, the higher the resolution of the trace.** This will allow you to make more accurate measurements of your signal and catch any errors that may exist.

What is the Right Sample Rate?

Harry Nyquist theorized that samples must have even spacing, and the sample rate should be at least twice the highest frequency of the signal. In other words:

$$F_{\text{sampling}} \geq 2F_{\text{signal}}$$

Though this theorem is valid in some situations, sometimes it's more complicated to find the best sample rate.

This 2x rule is relative to the Nyquist frequency, which is the point to which signals sample without aliasing. Many vendors will specify their oscilloscope bandwidth at the Nyquist frequency, which is technically correct, but there is a chance you aren't able to see some of the higher frequency components that exist in your signal.

Most oscilloscopes with a bandwidth of 1 GHz and below will have what is known as a Gaussian frequency response; see Figure 6. Because of the “roll-off” nature of the Gaussian response, some frequencies above the specified bandwidth can still be sampled, even though they are significantly attenuated. This is represented by the dark grey area in Figure 6. This makes it possible to gain more insight into your signal details. Note that the Nyquist frequency and bandwidth are specified at the same value in this figure.

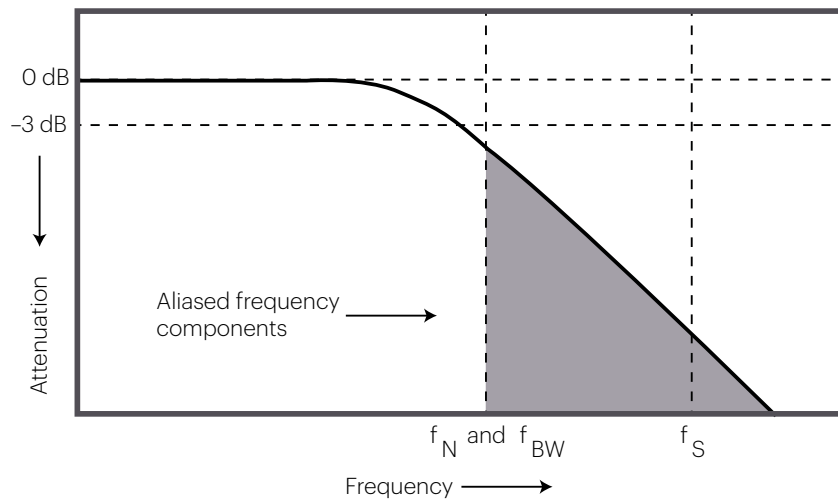


Figure 6. Graphical display showing Nyquist frequency equal to bandwidth

Figure 6 illustrates how specifying the Nyquist frequency equal to the bandwidth violates the Nyquist theorem. The theorem states that the highest frequency component sampled must be less than half the sampling frequency. You can see by the shaded area in Figure 6 that the highest possible frequency components exceed the sampling frequency (f_s).

Figure 7 shows an example of what your signal may look like if the Nyquist theorem is followed verbatim in a Gaussian response oscilloscope. The oscilloscope used to capture this signal has 500 MHz of bandwidth and a 1 GSa/s sample rate. Although the fundamental clock rate of this signal is within the Nyquist frequency and bandwidth, the edges occur much faster, well beyond the Nyquist frequency. This means that aliasing will occur, because the sampling frequency is not fast enough to accurately capture those edges. This affects the rise and fall time measurements you make.

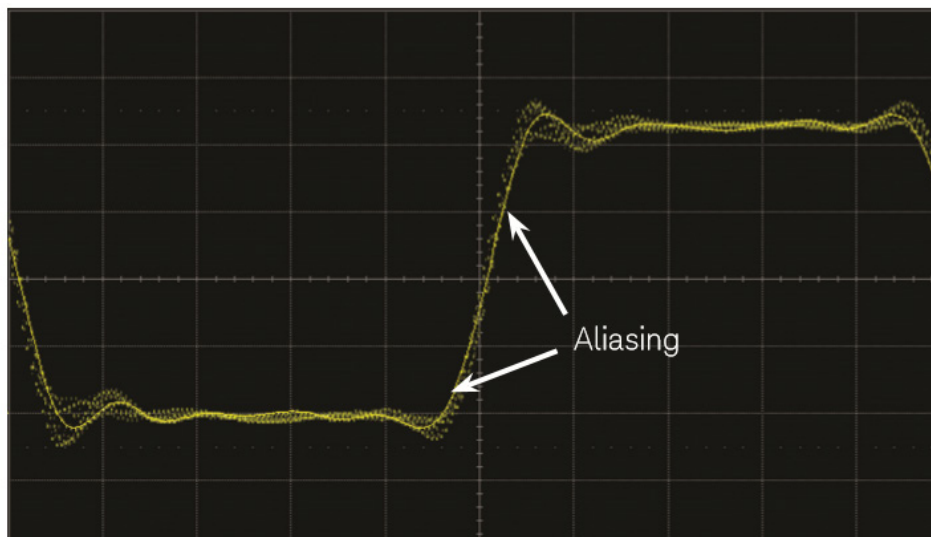


Figure 7. 500 MHz bandwidth scope sampling at 1 GSa/s produces aliased edges

With that in mind, oscilloscope vendors like Keysight have started designing their Gaussian response oscilloscopes to have a sample rate closer to 4x or 5x the bandwidth. This has proven to provide an improved level of signal detail with those faster edges.

In Figure 8, using the 4x multiplier rather than 2x, the sampling frequency is now significantly higher than twice the highest possible frequency component.

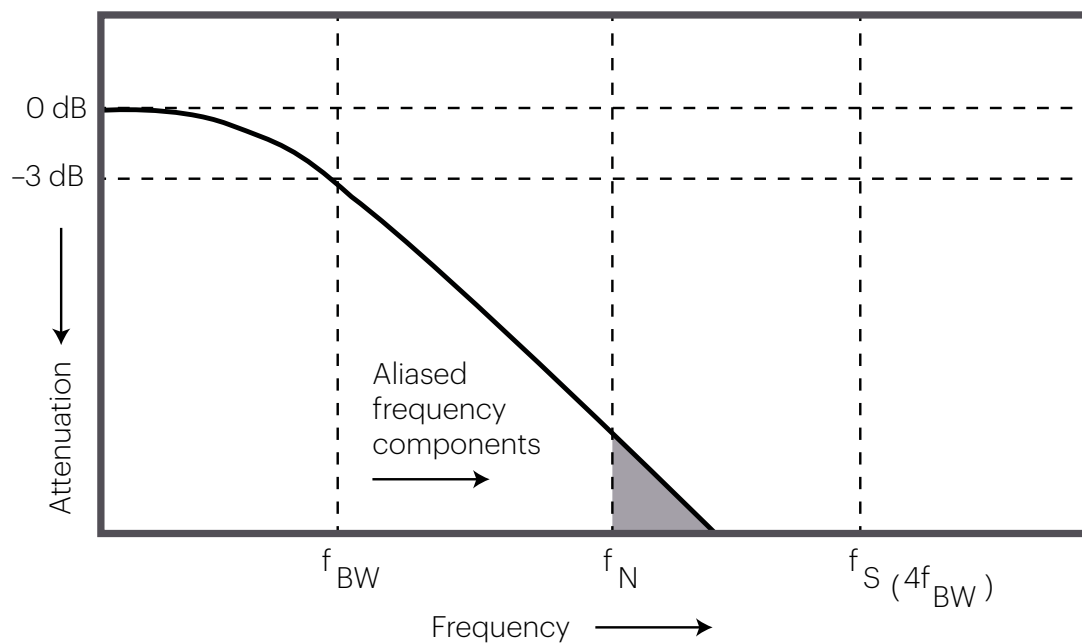


Figure 8. The graph illustrates sample frequency 4x the bandwidth frequency

It is important to ensure that your sample rate is high enough to capture that critical signal detail. Any additional bit of insight you can gain is helpful in testing.

When using oscilloscopes with a bandwidth of 1 GHz or higher, a sample rate that is around 2.5x higher than the bandwidth is typically adequate.

Also, keep in mind when multiple oscilloscope channels are turned on, the sample rate is usually split between these channels. Make sure your sample rate is fast enough when divided among channels. This information is available in the data sheet.

Acquisition Modes

To be sure you aren't missing anything in your analysis, it's important to understand your signals' strengths and weaknesses using various acquisition modes. Acquisition modes vary your oscilloscope's sampling method to analyze different signal characteristics. There are four modes:

1. **Normal mode:** Normal mode is the default mode and is the most commonly used mode for day-to-day measurements. It acquires samples at a specified sample rate and displays all of them on screen at each trigger event. This is the safest mode to use because there are no significant caveats to it.
2. **Averaging mode:** This mode captures multiple waveforms and averages them together. This is great for measuring periodic signals like a clock, or anything with a stable trigger. It is mainly used to hide transient noise or glitches to see the true, underlying signal. However, this mode should not be used for general debugging for that exact reason, only to get a glance at your true signal.

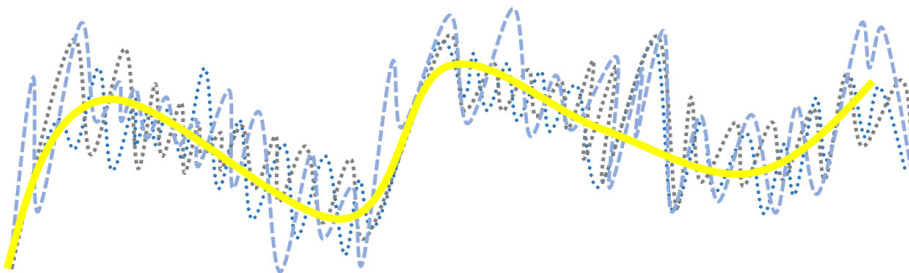


Figure 9. Averaging mode averages the values over multiple acquisitions

3. High-resolution mode (Hi-Res): This is another averaging mode. However, instead of waveform-to-waveform averaging, this performs point-to-point averaging. This allows you to capture glitches, aperiodic signals, and unstable triggers while still reducing some of the random noise riding on the signal. There is a higher effective number of bits as well since the noise is slightly reduced.

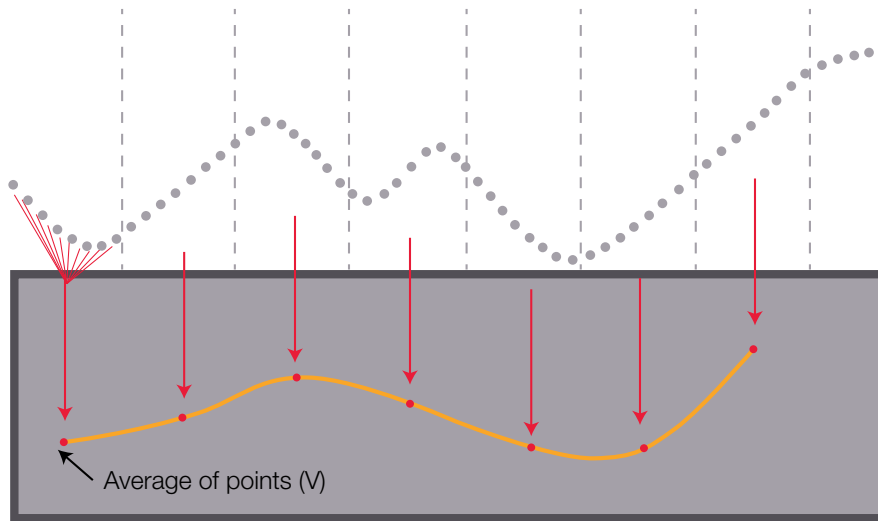


Figure 10. Hi-res mode averages the values from each interval

4. Peak detect mode: This mode takes the highest voltage sample value from a given interval and the lowest from the next, in alternating succession; see Figure 11. This helps you to see the extreme high and low values in your signal and catch any spikes or glitches that might otherwise be lost in the data compression process.

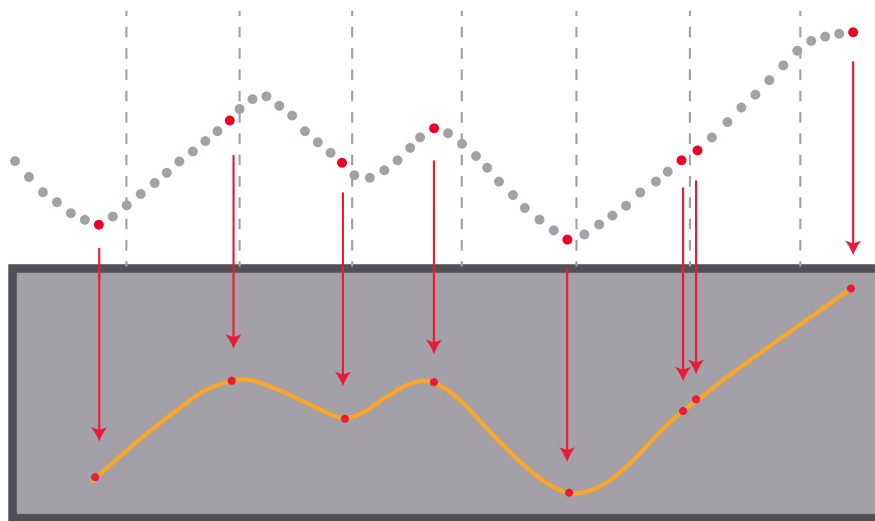


Figure 11. Peak detect mode takes the highest and lowest values in alternating order

Depending on what it is you are trying to view in your signal, acquisition modes are exceptionally beneficial in looking at the signal from a different perspective. You may find errors you didn't know were there, or you may discover something new about your true waveform when you average out the noise.

Avoiding Interleave Distortion

Some oscilloscopes can reach incredibly high sample rates, such as 10 GSa/s and higher. However, to achieve these rates, they must use two or more ADCs, synchronized with clock phase delays in a “one fires then the other fires” pattern. This effectively doubles the maximum sample rate.

At a high level, this technique seems useful. After all, the faster the sample rate, the higher the resolution. However, there is a catch.

If we go back to Nyquist's theorem, it states that the samples must all be equally spaced in time to produce an accurate trace. While this is relatively easy to do with a single ADC, it is much more difficult to get the phase delay clocks of the multiple ADCs perfectly synchronized. To achieve this, the synchronized ADCs must have the same vertical gain, offset, and frequency response.

If these interleaving requirements are not satisfied, the phase delay clocks will not be aligned causing inconsistent spacing between samples. In Figure 12, you can see how the two misaligned clocks can result in uneven spacing between samples.

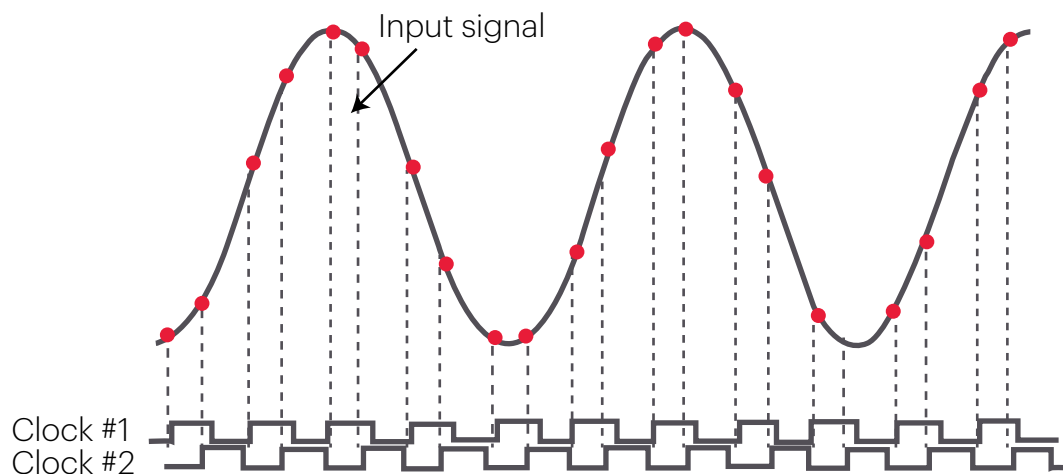


Figure 12. Uneven spacing between samples of an analog signal due to poorly interleaved ADCs

The processor in the oscilloscope assumes that all samples are collected at a consistent rate. This means if samples are collected at uneven intervals, when the waveform is digitally reconstructed it will appear distorted; see Figure 13. Much like the distorted signal with too low of a sample rate, any measurements you make on a signal like this will be completely inaccurate.



Figure 13. A 200 MHz sine wave captured on a 1 GHz scope at 10 GSa/s with interleaving is reconstructed incorrectly and shows distortion

In Figure 13 you can see the higher sample rate produces a less accurate waveform. Typically, a higher sample rate is preferred, but when improper interleaving technology enters the process, it brings enough inaccuracy to negate this method of achieving a faster sample rate. This is especially true when working with extremely high-frequency signals since even the smallest misalignment in clocking has a significant effect on the waveform.

However, there are many cases where interleaving is implemented correctly, like with the **Keysight S-Series oscilloscopes**. Interleaving isn't something to be afraid of, just something to be cautious of. There are also some extremely precise oscilloscopes, like the **Keysight UXR-Series**, that can reach incredibly high sample rates without interleaving ADCs. This will, of course, give you even more accuracy and higher signal integrity.

Aside from visually seeing the distortion in the trace on the screen, there is a test you can perform to see if interleave distortion exists in your oscilloscope and if it is responsible for any inaccuracies.

Fast Fourier Transform Test

When you see distortion on your signal, you can only guess at whether the distortion is caused by ADC interleaving or if there is another reason. Use a fast Fourier transform (FFT) to quickly and easily evaluate the nature of the distortion.

With an FFT, you can see the various frequency components of your signal. You should be able to look at the harmonics of your signal, but any irregular spikes found between these harmonics could be identified as harmonic distortion, noise, or interleave distortion.

In Figure 14, you can see an FFT display of a sine wave input. The fundamental frequency stands out, and you can look at the harmonics at consistent intervals. There are no unexpected frequency components between any of the harmonics or fundamental frequency, so this signal is not likely subject to interleave distortion.

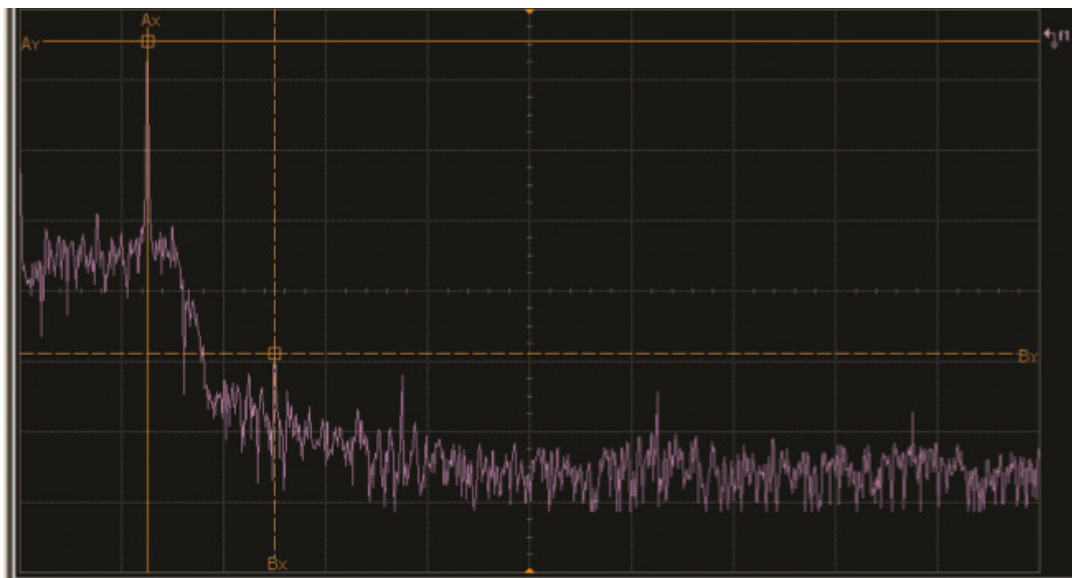


Figure 14. FFT of a sine wave input showing no distortion between harmonics

In Figure 15 you can see an FFT with multiple significant spurs outside the fundamental frequency and harmonics. In this case, interleaved ADCs are directly responsible for distortion.

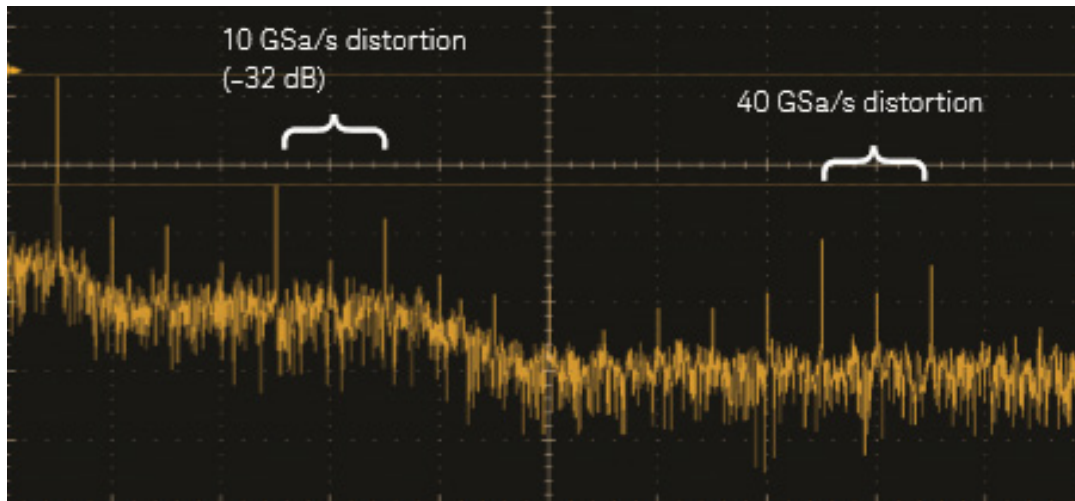


Figure 15. FFT shows spurs caused by phase clocking error between ADCs

It is not uncommon for scopes with extremely high sample rates to produce distorted signal reconstructions on screen. It is important to be aware of the possible downsides of faster sample rates and always check for those in any oscilloscope you are using or looking to purchase.

Key Takeaways

In summary, always be cautious when you are working with both low and high sample rates. This is a key specification to monitor when making measurements. Remember that the sample rate can automatically change as you adjust the time base settings, so be sure you are using the appropriate settings for the measurements you need to make.

The key things to remember about sample rate in your testing are:

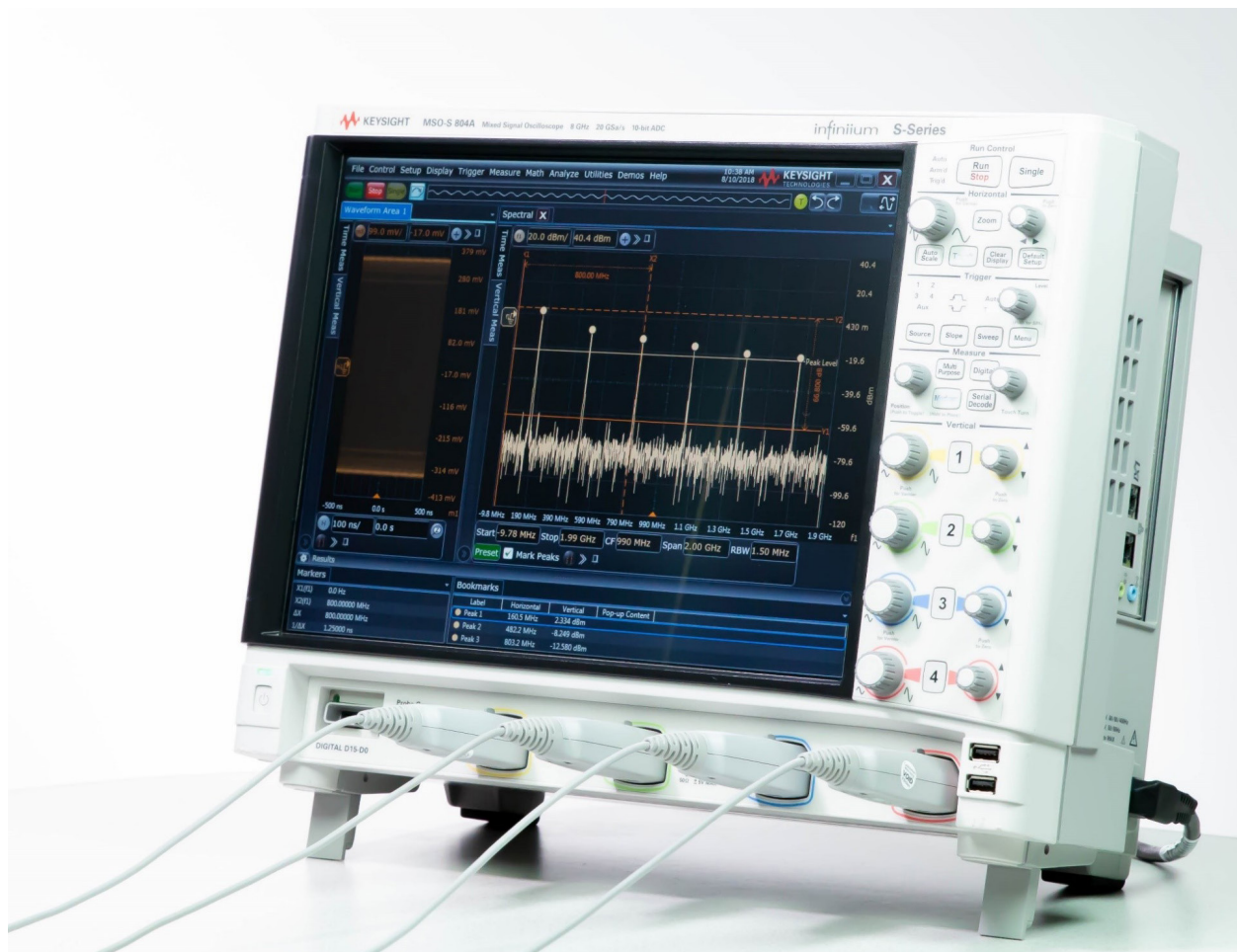
1. Sample rate refers to the rate at which the oscilloscope can collect data points, in Sa/s
2. Too low of a sample rate can cause distortion and skew your measurements
3. Acquisition modes will help you view your signal in different ways
4. Interleaved ADCs will present an impressive sample rate specification; however, if they are not properly synchronized this method can distort your signal

Conclusion

The sample rate is more than just an internal working of your oscilloscope. It has a profound impact on the trace you see. Sample rate determines whether you see an accurate representation of your signal, or a distorted one. As an engineer creating new digital designs in a world of increasingly fast data rates and high-precision devices, you don't have any room for unnecessary development delays. Don't let your oscilloscope's internal errors affect your measurements.

In addition to sample rate, there are a few other key considerations to ensure your oscilloscope gives you the highest signal integrity and best measurement accuracy.

To learn about the other important aspects of signal integrity, download the [How to Determine Oscilloscope Signal Integrity](#) eBook.



See a True Representation of Your Signal

Whether you are debugging your latest design, verifying compliance against an industry standard, or decoding a serial bus, it is critical that your oscilloscope displays a true representation of your signal. This requires world-class signal integrity, and Infiniium S-Series oscilloscopes were designed with that in mind. The superior time base, front end, and ADC technology blocks provide a platform with extremely low noise, low jitter, and high effective number of bits, giving you visibility into the true performance of your device.

Check out more details on the **S-Series'** signal integrity and measurement performance.

Need more bandwidth and even better performance? Check out these higher-bandwidth oscilloscopes:

- **V-Series (8 GHz to 33 GHz)**
- **Z-Series (20 GHz to 63 GHz)**
- **UXR-Series (13 GHz to 110 GHz)**